SPINE · NEET-SS / FET / INI-SS

Spinal tuberculosis and infection

1. Spinal tuberculosis (Pott's disease)

Epidemiology and pathology

- **Spine is the commonest site of skeletal TB (about 50%)**; India carries the largest TB burden. **Thoracolumbar junction (lower thoracic/upper lumbar) commonest**; **cervical** less common but more dangerous.
- **Haematogenous spread** via **Batson's valveless vertebral venous plexus** or arterial supply.
- **Patterns of vertebral involvement:**
  - **Paradiscal (commonest, about 80%)**: metaphyseal regions on either side of a disc (the disc is destroyed later than in pyogenic infection: **disc relatively preserved early**, then narrowed).
  - **Central (vertebral body)**: concertina collapse, mimics tumour.
  - **Anterior (subperiosteal)**: **anterior scalloping** of several vertebrae beneath the ALL (abscess tracking).
  - **Appendicular/posterior** (posterior elements): less common; early neurological deficit.
- **Cold abscess** tracks along fascial planes: **psoas abscess** (groin/thigh swelling), retropharyngeal (cervical: dysphagia, stridor), paravertebral (**"bird nest" / fusiform paravertebral shadow** on chest radiograph), lumbar triangle (Petit), **skip lesions** in about 5-15% (**MRI the whole spine**).
- **Pathological feature: caseating epithelioid granulomas** with Langhans giant cells.

Clinical features

- Insidious **back pain**, constitutional symptoms (**evening fever, night sweats, weight loss, anorexia**), **stiffness**, **gibbus/knuckle deformity** (localised kyphosis), cold abscess, **paraplegia (Pott's paraplegia)**.
- **Children:** worse kyphosis progression (growth); **"spine at risk" signs (Rajasekaran)**: **separation of facets**, **posterior retropulsion**, **lateral translation**, **toppling** (tilt of the upper vertebra). **Two or more signs** predict severe progression of kyphosis: consider surgery.

Pott's paraplegia

- **Early-onset (active disease, within about 2 years):** compression by **abscess, granulation tissue, caseous material, sequestra/disc**, or (rarely) **ischaemia/infective thrombosis** of the anterior spinal artery. **Good prognosis** with treatment.
- **Late-onset (healed disease, years later): internal gibbus (bony ridge)**, fibrosis/meningeal scarring, cord atrophy, reactivation. **Poor prognosis**.
- **Tuli's grading** (and **Jain's** modification) of neurological deficit (grades I-IV: negligible to complete paraplegia with sensory/sphincter loss).

Investigations

- **ESR, CRP** raised (non-specific), lymphocytosis; **Mantoux/IGRA** (supportive, not diagnostic in endemic areas).
- **Radiographs:** loss of paradiscal margins, disc narrowing (late), collapse, kyphosis, paravertebral shadow; **changes appear late (about 30-40% bone loss)**.
- **MRI (investigation of choice):** early marrow oedema, **paradiscal involvement with relative disc preservation**, **subligamentous spread across multiple levels**, **large paraspinal/psoas abscess with thin smooth enhancing wall**, skip lesions. (Pyogenic: disc destroyed early, small thick-walled abscess.)

- **CT-guided biopsy** for histology, **AFB smear, culture (liquid MGIT ~2-6 weeks)** and **CBNAAT/GeneXpert MTB/RIF** (rapid; **rifampicin resistance**; **Xpert Ultra** more sensitive). **Drug susceptibility testing** (MDR-TB rising).

Treatment

- **Antitubercular therapy (India NTEP/WHO): 2 months intensive phase (HRZE: isoniazid, rifampicin, pyrazinamide, ethambutol) + continuation phase HRE (WHO: HR)** for a **total of about 9-12 months for spinal TB** (extended to 18 months by some for slow responders). Pyridoxine with isoniazid. **MDR-TB:** longer regimens (bedaquiline-based, 18-24 months or newer shorter all-oral regimens).
- **Drug toxicity: isoniazid (hepatitis, peripheral neuropathy)**, **rifampicin (hepatitis, orange urine, enzyme induction: oral contraceptives)**, **pyrazinamide (hepatitis, hyperuricaemia/gout)**, **ethambutol (optic neuritis: red-green colour vision)**, streptomycin (ototoxicity, nephrotoxicity).
- **"Middle path regimen" (Tuli): ATT + rest/bracing**; **surgery for specific indications** or failure.
- **Surgical indications:**
  - **Neurological deficit** not improving (or worsening) on ATT, or **severe/rapidly progressive** deficit; **paraplegia in flexion**, **sphincter involvement**, **late-onset paraplegia**.
  - **Instability** and **progressive kyphosis** (children with spine-at-risk signs; **adult kyphosis over about 60 degrees**: risk of late paraplegia).
  - **Large abscesses** not responding (percutaneous drainage often suffices), **diagnostic uncertainty**, **failure of ATT/drug resistance**, **posterior element disease** with compression.
- **Surgery types:**

  - **Anterior radical debridement and strut grafting ("Hong Kong operation", Hodgson)**: historical gold standard.
  - **Posterior or posterolateral (transpedicular/costotransversectomy) debridement with pedicle screw instrumentation and fusion**: now commonly used (single posterior approach can decompress, correct kyphosis and stabilise; titanium implants are safe in TB).
  - **Anterolateral decompression** (Tuli; costotransversectomy, lateral extracavitary).
  - **Deformity correction:** posterior vertebral column resection/closing-opening wedge osteotomies for established kyphosis.
- **Medical Research Council (MRC) trials:** ambulatory chemotherapy with/without bracing was effective; **radical debridement + strut grafting** gave less kyphosis and faster fusion (Hong Kong).

2. Pyogenic spondylodiscitis (vertebral osteomyelitis)

- **Haematogenous** (adults over 50, diabetes, IV drug use, dialysis, immunosuppression, endocarditis) or **post-operative**. **Staphylococcus aureus commonest** (MRSA); gram-negatives (**E. coli** after urinary infection), **Pseudomonas** (IV drug users), streptococci (endocarditis association: **echocardiography** if bacteraemic), *Brucella* (endemic, Middle East/India: farm/raw milk; **Pedro Pons sign** anterosuperior endplate erosion; doxycycline + rifampicin/streptomycin), fungal (immunocompromised).
- **Adults: lumbar commonest**; infection starts in the **subchondral endplate** (segmental arteries) and spreads to the **disc early** (adult disc avascular; children's disc is vascular: primary discitis).
- **Features:** severe back pain, fever (about 50%), raised **CRP/ESR** (CRP used to monitor), **blood cultures positive in about 50-70%**, neurological deficit with **epidural abscess**.

- **MRI: disc T2 hyperintensity and enhancement**, **endplate erosion on both sides of the disc**, marrow oedema, paraspinal/epidural abscess.
- **Diagnosis: blood cultures**; if negative, **CT-guided biopsy before antibiotics** (if the patient is stable and not septic) for culture and histology (TB, fungal, malignancy).
- **Treatment:** targeted IV then oral antibiotics, **total about 6 weeks** (**Bernard et al., Lancet 2015: 6 weeks as effective as 12 weeks** for pyogenic spondylodiscitis), bracing for pain. **Surgery: neurological deficit, epidural abscess with compression**, **instability/deformity**, **failure of medical treatment**, need for diagnosis. **Debridement + instrumentation and fusion** (titanium and even pedicle screws are safe in active infection).

3. Spinal epidural abscess

- **Risk factors:** diabetes, IV drug use, alcohol, immunosuppression, recent spinal procedure, bacteraemia. **S. aureus** (about 60-70%).
- **Classic triad: back pain, fever, neurological deficit** (present in under 20% at presentation). Progression: **back pain, radicular pain, weakness, paralysis**.
- **MRI with gadolinium** (rim-enhancing collection) of the whole spine.
- **Treatment: urgent surgical decompression and drainage + antibiotics** for **neurological deficit** (prognosis depends on pre-operative neurology and duration of paralysis: **paralysis over 24-36 hours rarely recovers**). **Antibiotics alone** only for **neurologically intact** patients without risk factors for failure (close monitoring; **failure predictors: diabetes, MRSA, CRP over 115, WBC over 12,500, positive blood cultures, age over 65**) or patients unfit for surgery/complete paralysis over 72 hours.

4. Post-operative spinal infection

- **Surgical site infection about 1-4%** (higher with instrumentation, posterior approach, long fusions, obesity, diabetes, neuromuscular scoliosis, trauma, prolonged surgery, transfusion).
- **Early (within about 4-6 weeks): debridement, irrigation, retain stable implants**, cultures, IV antibiotics, ± vacuum dressing; **vancomycin powder** in the wound reduces SSI in some studies (prophylaxis).
- **Late/delayed infection:** low-virulence organisms (***Cutibacterium acnes***, coagulase-negative staphylococci); **remove implants if fusion is solid**; keep if not fused and re-instrument as needed.

5. Paediatric discitis

- **Ages 1-5** (vascular channels in the disc), toddlers **refusing to walk, sit or bend**, back or abdominal pain, mild fever; **S. aureus/Kingella**. **MRI**. Treatment: **antibiotics (anti-staphylococcal) and rest/brace**; biopsy rarely needed.

6. Differentiating TB vs pyogenic vs tumour

Feature	TB	Pyogenic	Metastasis/tumour
Onset	**Insidious (months)**	**Acute (days-weeks)**	Gradual, night pain
Disc	**Preserved early, subligamentous spread over several levels**	**Destroyed early**	**Spared**
Abscess	**Large, thin smooth wall, calcification**	Smaller, thick irregular wall	Soft-tissue mass

Feature	TB	Pyogenic	Metastasis/tumour
Posterior elements	Sometimes	Rarely	**Pedicle commonly involved** (winking owl sign)
Skip lesions	Common	Rare	Multiple levels possible
Deformity	**Gibbus/kyphosis**	Less	Pathological collapse

Exam pearl: **Spinal TB: thoracolumbar, paradiscal commonest, disc preserved early, large thin-walled abscess, skip lesions; CBNAAT gives rifampicin resistance.** ATT about 9-12 months; surgery for neurological deficit not improving, instability, progressive kyphosis (children: spine-at-risk signs, 2 or more = surgery).

Exam pearl: **Early-onset Pott's paraplegia (active disease) has a good prognosis; late-onset (healed, internal gibbus) poor. Ethambutol = optic neuritis; INH = neuropathy (give pyridoxine).**

Exam pearl: **Pyogenic spondylodiscitis: S. aureus, disc destroyed early; blood cultures then CT-guided biopsy before antibiotics; 6 weeks of antibiotics (Bernard). Epidural abscess with deficit = urgent decompression.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.